

LOVOZERO, Russian Federation

WMO: 221270

Lat: 68.004N

Lon: 35.031E

Elev: 531

StdP: 14.42

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.0	-24.0	-34.3	0.8	-29.9	-28.2	1.2	-24.0	24.4	16.6	21.4	20.9	0.4	40	0.736

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	76.2	62.5	71.8	60.2	68.0	58.2	64.3	73.4	61.8	69.6	59.4	66.2	6.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	80.8	67.5	58.2	73.9	65.6	56.0	68.0	63.1	29.8	73.5	27.9	70.0	26.3	66.3	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	16.2	14.0	DB	-35.2	80.6	6.0	4.0	-39.5	83.5	-43.0	85.8	-46.3	88.0	-50.6	90.9	(4)
(5)			WB	-35.3	66.8	5.9	3.1	-39.5	69.0	-43.0	70.8	-46.3	72.6	-50.6	74.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.6	8.2	7.3	16.8	28.3	38.6	48.9	55.9	51.8	43.6	32.3	19.9	14.0		
	DBStd	19.35	14.37	15.00	11.26	8.35	7.67	7.84	7.29	6.41	6.03	7.74	11.19	12.53		
	HDD50	7497	1295	1195	1030	650	361	114	25	52	208	549	902	1116		
	HDD65	12567	1760	1615	1495	1100	817	485	293	412	643	1014	1352	1581		
	CDD50	423	0	0	0	0	9	82	209	108	15	0	0	0		
	CDD65	19	0	0	0	0	0	3	12	4	0	0	0	0		
	CDH74	191	0	0	0	0	2	32	117	40	0	0	0	0		
	CDH80	18	0	0	0	0	0	3	11	4	0	0	0	0		
	WSAvg	5.7	5.2	5.5	6.1	6.4	6.8	6.4	5.7	5.1	5.4	5.5	5.1	5.5		
	PrecAvg	19.5	1.0	1.0	1.0	1.0	1.5	2.4	3.0	2.4	2.0	1.9	1.2	1.2		
	PrecMax	26.5	1.5	1.7	3.1	2.6	2.9	4.4	4.5	4.5	3.7	3.7	2.6	2.1		
(15) Precipitation	PrecMin	15.9	0.2	0.2	0.5	0.2	0.7	0.7	1.2	0.9	0.1	0.4	0.3	0.4		
	PrecStd	2.6	0.3	0.4	0.6	0.5	0.6	1.2	0.8	0.8	1.1	0.9	0.6	0.5		
	DB	36.5	37.4	42.3	51.2	69.1	78.9	81.4	78.7	66.1	51.9	41.6	38.9			
	MCWB	33.5	33.5	36.0	42.6	54.0	62.3	66.0	64.5	56.1	47.3	38.8	35.3			
	DB	32.4	34.1	37.8	46.2	61.7	72.0	77.3	72.7	61.5	48.0	37.7	35.9			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	30.5	31.5	33.3	39.7	50.4	58.4	63.2	61.4	53.7	44.5	35.3	33.1			
	DB	29.3	31.1	34.7	42.7	56.9	67.6	73.8	67.6	57.7	45.1	35.3	33.0			
	MCWB	27.7	29.0	31.2	37.1	47.3	56.2	62.0	59.0	52.0	42.7	33.5	31.2			
	DB	26.3	27.9	32.0	39.7	52.3	63.5	69.9	63.5	53.8	42.6	33.1	30.3			
	MCWB	25.1	26.3	29.4	35.3	44.7	53.7	60.2	56.8	49.8	40.6	31.9	28.9			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	34.1	34.7	36.6	43.4	55.5	63.8	68.4	65.8	57.4	47.6	39.5	35.7			
	MCDB	36.0	37.0	41.3	50.2	67.5	75.9	77.4	76.5	63.1	50.7	41.2	38.3			
	WB	30.9	31.7	33.7	40.0	51.1	60.2	65.6	62.6	54.6	44.9	35.7	33.4			
	MCDB	32.4	33.9	37.1	45.5	60.6	69.8	74.5	70.3	60.1	47.1	37.2	35.4			
	WB	27.8	29.0	31.5	37.6	47.9	57.3	63.2	60.0	52.3	43.0	33.7	31.3			
(33) Mean Daily Temperature Range	MCDB	29.1	31.0	34.1	41.9	55.6	66.0	71.2	66.2	57.0	45.1	35.0	33.0			
	WB	25.1	26.3	29.5	35.7	45.0	54.7	61.0	57.6	50.2	40.8	32.1	28.9			
	MCDB	26.2	27.9	32.1	39.1	51.9	62.6	68.3	62.7	53.6	42.4	33.0	30.3			
	MDBR	15.8	16.7	17.2	14.4	13.6	15.2	15.9	14.3	12.8	9.0	11.4	14.8			
	MCDBR	14.6	14.8	16.0	16.7	23.7	22.5	23.8	22.5	21.3	12.1	8.9	12.2			
(40) Clear-Sky Solar Irradiance	MCWBR	13.3	13.0	12.8	11.2	13.9	11.6	11.7	12.5	13.9	9.3	7.9	10.8			
	MCDBR	13.7	14.4	14.1	15.7	22.4	19.9	20.9	19.4	18.7	11.3	8.4	12.0			
	MCWBR	12.8	13.0	11.7	10.8	13.7	11.3	11.2	11.5	12.9	9.0	7.7	10.7			
	taub	0.209	0.262	0.303	0.335	0.348	0.347	0.368	0.347	0.319	0.281	N/A	N/A			
	taud	1.902	2.036	2.079	2.126	2.318	2.454	2.418	2.450	2.490	2.237	N/A	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	26	163	226	254	267	272	260	253	232	160	N/A	0			
	Edh at Noon	3	16	28	36	33	30	30	26	19	13	N/A	0			
	RadAvg	11	152	605	1176	1447	1540	1435	980	543	165	28	0			
	RadStd	1	18	36	75	155	154	192	96	56	25	3	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page